

UBUNTU

Сети

```
vboxuser@ubuntu1:~/cache-lab$ hostname
ubuntu1
vboxuser@ubuntu1:~/cache-lab$ ip -br a
lo                UNKNOWN      127.0.0.1/8 ::1/128
enp0s3            UP           10.0.2.15/24 fd17:625c:f037:2:c0f5:1430:7589:984d/64 fd17:625c:f037:2:a00:27ff:fea4:af6
8/64 fe80::a00:27ff:fea4:af68/64
enp0s8            UP           192.168.56.10/24 fe80::a00:27ff:fe51:c1e4/64
vboxuser@ubuntu1:~/cache-lab$
```

```
vboxuser@ubuntu1:~/cache-lab$ sudo ufw status numbered
[sudo] password for vboxuser:
Status: active
```

	To	Action	From
	--	-----	----
[1]	OpenSSH	ALLOW IN	Anywhere
[2]	22/tcp	ALLOW IN	Anywhere
[3]	8080/tcp	ALLOW IN	192.168.56.20
[4]	8080/tcp	DENY IN	Anywhere
[5]	5432/tcp	DENY IN	Anywhere
[6]	OpenSSH (v6)	ALLOW IN	Anywhere (v6)
[7]	22/tcp (v6)	ALLOW IN	Anywhere (v6)
[8]	8080/tcp (v6)	DENY IN	Anywhere (v6)
[9]	5432/tcp (v6)	DENY IN	Anywhere (v6)

```
vboxuser@ubuntu1:~/cache-lab$
```

Сервисы

```
vboxuser@ubuntu1:~/cache-lab$ systemctl status postgresql
● postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/usr/lib/systemd/system/postgresql.service; enabled; preset: enabled)
   Active: active (exited) since Tue 2026-09-22 07:38:39 UTC; 3h 42min ago
   Process: 1643 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 1643 (code=exited, status=0/SUCCESS)
      CPU: 2ms

Sep 22 07:38:39 ubuntu1 systemd[1]: Starting postgresql.service - PostgreSQL RDBMS...
Sep 22 07:38:39 ubuntu1 systemd[1]: Finished postgresql.service - PostgreSQL RDBMS.
vboxuser@ubuntu1:~/cache-lab$ systemctl status backend-api
● backend-api.service - Backend API
   Loaded: loaded (/usr/lib/systemd/system/backend-api.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-09-22 07:38:39 UTC; 3h 42min ago
   Main PID: 1650 (backend-api)
      Tasks: 6 (limit: 14775)
     Memory: 11.5M (peak: 12.0M)
        CPU: 70ms
    CGroup: /system.slice/backend-api.service
            └─1650 /usr/local/bin/backend-api

Sep 22 07:38:39 ubuntu1 systemd[1]: Started backend-api.service - Backend API.
Sep 22 07:38:39 ubuntu1 backend-api[1650]: 2026/09/22 07:38:39 Backend API запущен на :8080
```

Запросы

```
vboxuser@ubuntu1:~/cache-lab$ curl "http://127.0.0.1:8080/user?id=5"
{"id":5,"name":"Charlie Davis","age":35}
vboxuser@ubuntu1:~/cache-lab$ pg_isready -h 127.0.0.1 -p 5432
127.0.0.1:5432 - accepting connections
vboxuser@ubuntu1:~/cache-lab$
```

CENTOS

Сети

```
gleb@localhost:~$ ip -br a
lo                UNKNOWN          127.0.0.1/8 ::1/128
enp0s3            UP                10.0.2.15/24 fd17:625c:f037:2:a00:27ff:fe01:e554/64 fe80::a00:27ff:fe01:e554/64
enp0s8            UP                192.168.56.20/24 fe80::2999:e97a:1b28:9c3c/64
gleb@localhost:~$ sudo firewall-cmd --zone=public --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3 enp0s8
  sources:
  services: cockpit dhcpv6-client ssh
  ports: 5000/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
gleb@localhost:~$
```

Сервисы

```
gleb@localhost:~$ systemctl status redis
● redis.service - Redis In-Memory Data Store
   Loaded: loaded (/etc/systemd/system/redis.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-09-22 14:36:01 MSK; 4min 7s ago
  Invocation: b9df6e6d29f3414aa32c1f25831d9d08
    Main PID: 1265 (redis-server)
      Tasks: 6 (limit: 100273)
     Memory: 23.8M (peak: 24.3M)
        CPU: 480ms
    CGroup: /system.slice/redis.service
            └─1265 /usr/local/bin/redis-server 127.0.0.1:6379

сен 22 14:36:01 localhost.localdomain redis-server[1265]: 1265:M 22 Sep 2026 14:36:01.454 * monotonic clock: POSIX clock_gettime
сен 22 14:36:01 localhost.localdomain redis-server[1265]: 1265:M 22 Sep 2026 14:36:01.455 # Failed to write PID file: Permission denied
сен 22 14:36:01 localhost.localdomain redis-server[1265]: 1265:M 22 Sep 2026 14:36:01.455 * Running mode=standalone, port=6379.
сен 22 14:36:01 localhost.localdomain redis-server[1265]: 1265:M 22 Sep 2026 14:36:01.455 * Server initialized
сен 22 14:36:01 localhost.localdomain redis-server[1265]: 1265:M 22 Sep 2026 14:36:01.457 * Loading RDB produced by version 7.4.11
сен 22 14:36:01 localhost.localdomain redis-server[1265]: 1265:M 22 Sep 2026 14:36:01.457 * RDB age 20 seconds
сен 22 14:36:01 localhost.localdomain redis-server[1265]: 1265:M 22 Sep 2026 14:36:01.457 * RDB memory usage when created 1.08 Mb
сен 22 14:36:01 localhost.localdomain redis-server[1265]: 1265:M 22 Sep 2026 14:36:01.457 * Done loading RDB, keys loaded: 0, keys expired: 0.
сен 22 14:36:01 localhost.localdomain redis-server[1265]: 1265:M 22 Sep 2026 14:36:01.457 * DB loaded from disk: 0.002 seconds
сен 22 14:36:01 localhost.localdomain redis-server[1265]: 1265:M 22 Sep 2026 14:36:01.457 * Ready to accept connections tcp
gleb@localhost:~$ systemctl status cache-api
● cache-api.service - Flask Cache API
   Loaded: loaded (/usr/lib/systemd/system/cache-api.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-09-22 14:36:01 MSK; 4min 19s ago
  Invocation: 963cc17800e3456f8a6eee9c5ec003ef
    Main PID: 1453 (python3)
      Tasks: 1 (limit: 100273)
     Memory: 43.8M (peak: 45.8M)
        CPU: 371ms
    CGroup: /system.slice/cache-api.service
            └─1453 /usr/bin/python3 /usr/local/lib/cache-api/cache-api.py

сен 22 14:36:01 localhost.localdomain systemd[1]: Started cache-api.service - Flask Cache API.
сен 22 14:36:02 localhost.localdomain python3[1453]: * Serving Flask app 'cache-api'
сен 22 14:36:02 localhost.localdomain python3[1453]: * Debug mode: off
сен 22 14:36:02 localhost.localdomain python3[1453]: WARNING: This is a development server. Do not use it in a production deployment. Use a production
сен 22 14:36:02 localhost.localdomain python3[1453]: * Running on all addresses (0.0.0.0)
сен 22 14:36:02 localhost.localdomain python3[1453]: * Running on http://127.0.0.1:5000
сен 22 14:36:02 localhost.localdomain python3[1453]: * Running on http://10.0.2.15:5000
сен 22 14:36:02 localhost.localdomain python3[1453]: Press CTRL+C to quit
lines 1-19/19 (END) ...skipping...
```

Запросы

```
gleb@localhost:~$ curl "http://192.168.56.10:8080/user?id=5"
{"id":5,"name":"Charlie Davis","age":35}
gleb@localhost:~$ curl "http://127.0.0.1:5000/user?id=5"
{"cached":false,"user":{"age":35,"id":5,"name":"Charlie Davis"}}
gleb@localhost:~$ curl "http://127.0.0.1:5000/user?id=5"
{"cached":true,"user":{"age":35,"id":5,"name":"Charlie Davis"}}
gleb@localhost:~$ redis-cli GET 5
"{\"id\":5,\"name\":\"Charlie Davis\",\"age\":35}\n"
gleb@localhost:~$
```

Пакеты

```
vboxuser@ubuntu1:~/cache-lab$ dpkg -s backend-api | grep -E 'Package|Version|Status'
Package: backend-api
Status: install ok installed
Version: 1.0.0
vboxuser@ubuntu1:~/cache-lab$
```

```
gleb@localhost:~$ rpm -q cache-api
cache-api-1.0.0-1.el10.x86_64
gleb@localhost:~$
```

```
PS C:\Users\glebm\OneDrive\Рабочий стол\Project> curl.exe "http://192.168.56.20:5000/user?id=5"
{"cached":true,"user":{"age":35,"id":5,"name":"Charlie Davis"}}
PS C:\Users\glebm\OneDrive\Рабочий стол\Project> Test-NetConnection 192.168.56.10 -Port 8080
ПРЕДУПРЕЖДЕНИЕ: TCP connect to (192.168.56.10 : 8080) failed
```

```
ComputerName      : 192.168.56.10
RemoteAddress     : 192.168.56.10
RemotePort        : 8080
InterfaceAlias    : Ethernet 4
SourceAddress     : 192.168.56.1
PingSucceeded     : True
PingReplyDetails (RTT) : 2 ms
TcpTestSucceeded  : False
```

```
PS C:\Users\glebm\OneDrive\Рабочий стол\Project> Test-NetConnection 192.168.56.20 -Port 6379
ПРЕДУПРЕЖДЕНИЕ: TCP connect to (192.168.56.20 : 6379) failed
```

```
ComputerName      : 192.168.56.20
RemoteAddress     : 192.168.56.20
RemotePort        : 6379
InterfaceAlias    : Ethernet 4
SourceAddress     : 192.168.56.1
PingSucceeded     : True
PingReplyDetails (RTT) : 0 ms
TcpTestSucceeded  : False
```

```
PS C:\Users\glebm\OneDrive\Рабочий стол\Project> Test-NetConnection 192.168.56.10 -Port 5432
ПРЕДУПРЕЖДЕНИЕ: TCP connect to (192.168.56.10 : 5432) failed
```

```
ComputerName      : 192.168.56.10
RemoteAddress     : 192.168.56.10
RemotePort        : 5432
InterfaceAlias    : Ethernet 4
SourceAddress     : 192.168.56.1
PingSucceeded     : True
PingReplyDetails (RTT) : 0 ms
TcpTestSucceeded  : False
```

```
PS C:\Users\glebm\OneDrive\Рабочий стол\Project> Test-NetConnection 192.168.56.20 -Port 5000
```

```
ComputerName      : 192.168.56.20
RemoteAddress     : 192.168.56.20
RemotePort        : 5000
InterfaceAlias    : Ethernet 4
SourceAddress     : 192.168.56.1
PingSucceeded     : True
PingReplyDetails (RTT) : 0 ms
TcpTestSucceeded  : True
```